

Analysis & Control of a Segway

Course Project

December 2024

In this project, your objective is to investigate the dynamics of a Segway and to develop appropriate feedback control systems. The Segway system is illustrated in a simplified, cartoon-style diagram, as shown in Fig. 1.

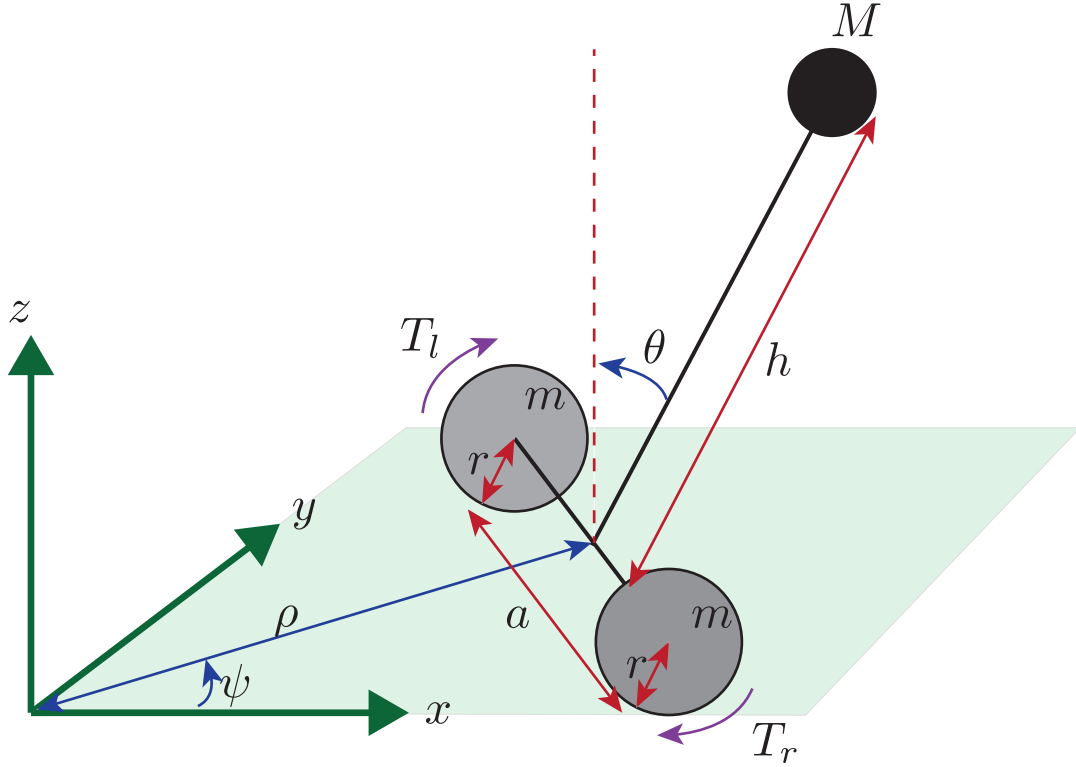


Figure 1: Segway Schematic.

The Segway has two identical wheels, each with a mass m , inertia I , and radius r . These wheels are linked by a massless axle of length a . The body of the Segway is attached to the axle's midpoint. The body's center of mass is at a height h above the axle when it is in the upright vertical configuration, and the body has a mass M which is influenced by gravitational force. The effective moments of inertia for the entire system are represented by I_x , I_y , and I_z . The Segway is designed to move in the xy -plane. This movement is characterized by no slipping or friction. The primary location variables are denoted as (ρ, ψ, θ) . Here, (ρ, ψ) represent the polar coordinates of the axle's midpoint in the xy -plane, while θ indicates the vertical angle of the Segway's body. The inputs consist of two torques applied to the wheels, labeled as T_l and T_r for the left and right wheels, respectively.

One can derive the equations of motion to obtain

$$\begin{bmatrix} M + 2m + \frac{2I}{r^2} & 0 & Mh \cos(\theta) \\ 0 & 2a^2m + (Mh^2 + I_x) \sin^2(\theta) + I_z \cos^2(\theta) + \frac{2Ia^2}{r^2} & 0 \\ Mh \cos(\theta) & 0 & Mh^2 + I_y \end{bmatrix} \begin{bmatrix} \ddot{\rho} \\ \ddot{\psi} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} \frac{T_l + T_r}{r} + Mh \sin(\theta) (\omega_1^2 + \omega_2^2) \\ \frac{a(T_r - T_l)}{r} - 2\omega_1\omega_2 \cos(\theta) \sin(\theta) (mh^2 + I_x - I_z) - Mh\omega_1 v \sin(\theta) \\ (Mh^2 + I_x - I_z) \omega_1^2 \cos(\theta) \sin(\theta) - T_l - T_r + Mgh \sin(\theta) \end{bmatrix}, \quad (1)$$

where $v := \dot{\rho}$, $\omega_1 := \dot{\psi}$ and $\omega_2 := \dot{\theta}$.

1. With reference to equation (1), what is the 6-dimensional state vector x of this system?
2. Create a MATLAB live script named “Linearization.mlx” that performs the following tasks using symbolic calculations:
 - (a) Develop a nonlinear state-space model to represent the dynamics of the system.
 - (b) Confirm that the state $\bar{x}$, where all its components are zero, is a fixed point when both T_l and T_r are set to zero. Identify an additional fixed point as well.
 - (c) Linearize the dynamics around the two identified fixed points. Specifically, calculate symbolically the matrices A and B for each fixed point.
3. For the remainder of the project, use the following designated parameter values:

m	0.141	kg
M	2.313	kg
h	0.254	m
a	0.165	m
r	0.0615	m
I	2.5e-4	kg.m ²
I_x	0.1986	kg.m ²
I_y	0.1942	kg.m ²
I_z	0.0056	kg.m ²
g	9.81	N.m ² /kg ²

Create a separate MATLAB script named “OpenLoop_Nonlinear_Simulation.m” that performs the following tasks:

- (a) Generate the torque inputs $T_l(t)$ and $T_r(t)$ as illustrated in Fig. 2. Use a time horizon of 10,000 steps spanning from $t = 0$ to $t = 20$ seconds. Display these inputs in a separate figure.

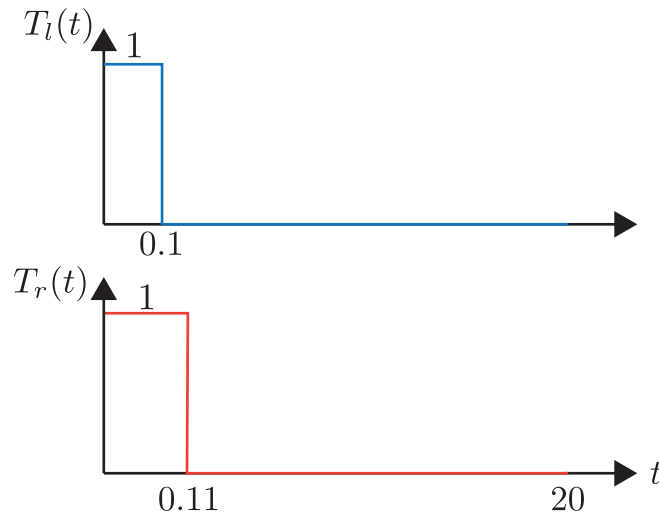


Figure 2: Inputs Profile.

- (b) Perform a nonlinear simulation to observe how all state-space variables react to these inputs. Present the response of each state variable on individual subplots within a single figure. *Hint: Instead of retyping the system matrices manually, you can copy the output from the live editor in 2(c). However, MATLAB may throw an error if the copied expression include, say for example, $\text{sym}(2)$. In this case, it suffices to remove the sym command and leave 2 as a double.*
 - (c) In a different figure, plot the Segway’s path by using the Cartesian coordinates calculated from $x(t) = \rho(t) \cos(\psi(t))$ and $y(t) = \rho(t) \sin(\psi(t))$. This should be plotted in the xy -plane.
 - (d) Analyze the results from your simulation to draw conclusions regarding the system’s stability.
4. Evaluate the stability of the two fixed points by computing the associated eigenvalues. To accomplish this, create a separate MATLAB script titled “Stability_Analysis”. Analyze the differences in stability types between these two fixed points and provide a mechanical explanation for these variations.

5. For the remainder of the project, consider only the fixed point where all components are zero. Assess whether the system is controllable around this fixed point by writing a separate MATLAB script named “Controllability.m”.
6. Create a separate MATLAB script titled “Pole_Placement.m” that accomplishes the following tasks:
 - (a) Calculate the full-state feedback gain necessary to place the closed-loop eigenvalues at $(-1, -2, -3, -4, -5, -6)$. Verify this by examining the eigenvalues of the resulting closed-loop system.
 - (b) Do simulations for both the nonlinear and linearized versions of the system under closed-loop control using the calculated full state feedback gain. Perform these simulations without any external inputs but starting from the initial condition

$$x_0 = \left(1; \frac{\pi}{4}; \theta_0; 0; 0; 0\right) \quad \text{for } \theta_0 = \pi/6, \pi/4, \pi/3. \quad (2)$$

The script should produce two figures similar to those in 3(b) and 3(c), displaying the states of both the linearized and nonlinear systems superimposed for comparison. In the second figure, overlay the path plots of the linearized and nonlinear closed-loop systems. Finally, analyze and discuss the effects of incrementally increasing θ_0 on the nonlinear system’s behavior as compared to the linearized system.

7. Develop a MATLAB script named “LQR.m” to revisit the tasks in sections 6(a) and (b), this time utilizing the optimal feedback gain derived from the Linear Quadratic Regulator (LQR). This script should:
 - (a) Implement the LQR approach to calculate the optimal feedback gain. This involves, selecting the Q and R matrices to achieve a desirable dynamic response, and applying the LQR algorithm to find the optimal gain.
 - (b) Provide a clear rationale for the selection of the Q and R matrices.
 - (c) Report where the LQR places the closed-loop eigenvalues, highlighting the differences in system behavior and stability compared to the original approach.
8. For the remaining part of the project, consider the initial condition where all components are zero except for $\theta(0) = \pi/6$. However, assume that this specific initial condition is unknown (set, the initial condition for the Kalman Filter is zero). Additionally, assume that your sensors, capable of measuring the location variables (ρ, ψ, θ) , are not free from noise and have an estimated noise power of 10^{-4} . The ultimate aim is to design a controller that can guide the Segway to a predefined setpoint location $(\rho_r, \psi_r) = (2, \frac{\pi}{4})$ while maintaining an upright position. Follow these steps:
 - (a) With the outputs being ρ, ψ and θ , check the observability of the system by creating a separate Matlab Script titled “Observability.m”.
 - (b) Create a MATLAB script titled “LQGI.Calculations.m”. This script should calculate the feedback gain incorporating an integrator to ensure convergence of $\rho(t)$ and $\psi(t)$ to their respective setpoint values, ρ_r and ψ_r .
 - (c) Construct a Simulink Model named “LQGI.Linearized.slx”. This model will simulate the LQG controller for the linearized plant, including the integrator, to achieve the setpoint. Incorporate sensor noise with the specified power. *Hint: Select appropriate C matrices for the Kalman Filter, LQI, and the plant, as each requires a different C matrix.* The Simulink model should have the following visualizations:
 - i. Six scopes, each representing a state variable, to overlay the real, non-noisy state against the Kalman Filter estimate. Each scope should display two signals: the actual state and the state estimated by the Kalman Filter. Adjust the parameters of the Kalman Filter to obtain accurate estimates.
 - ii. A scope to monitor the inputs applied to the plant.
 - iii. A new block named “XY Graph” used to plot the path in the xy -plane, calculating the Cartesian coordinates in a manner similar to section 3(c).
 - (d) After achieving satisfactory responses from the linearized model, create another Simulink Model named “LQGI.Nonlinear.slx”. This model should be the same as the linearized version, but with the linear plant replaced by the nonlinear subsystem provided to you as “Segway.slx”.

You are required to submit all the MATLAB and Simulink files referenced above, along with a concise write-up, not exceeding one page, addressing the specified questions. Please send these materials via email to maurice.flo@bsse.ethz.ch. Ensure that the MATLAB and Simulink files are fully operational and ready for execution.

Good luck!

